Data Science Lab Project

Life Satisfaction in OECD Countries

A Multidimensional Well-being Analysis with a Focus on Italy

Authors' Details

Family name	First name	Badge number	Email
Dawra	Tony	946852	t.dawra@campus.unimib.it
Cavallo	Davide	880593	d.cavallo4@campus.unimib.it
Villani	Giacomo	880819	g.villani7@campus.unimib.it

Keywords

Well-being; OECD; Life satisfaction; Social indicators; Italy

Name of the Dataset

OECD Well-being Data Monitor – Current well-being dataset, downloaded from the OECD Data Explorer.

Declaration on AI Tools

AI tools were used to help prepare reproducible R code for data cleaning, descriptive statistics, correlation analysis, visualization, construction of dimension scores, and simple regression modelling. The final data selection, execution of the analysis, verification of numerical outputs, and interpretation of the results, formulate the question of the research question, organization of the report, and the drafting of the explanatory text remain the responsibility of the authors. AI tools were not used to fabricate data, numerical results, or references.

Abstract

This study investigates which dimensions of well-being are most associated with life satisfaction across OECD countries, with a specific focus on Italy compared with the OECD average excluding Italy. The analysis uses the OECD Well-being Data Monitor, Current well-being dataset, covering 2010–2025, 38 OECD countries, and 21 selected measures. Since indicators are not always available for the same year, the latest available observation is retained for each country and measure. Life satisfaction is the outcome variable, while explanatory indicators are grouped into seven dimensions: economic security, work quality, housing, health, social connections, civic engagement, and environment and safety. The methodology combines descriptive statistics, missing-value analysis, correlations, standardized dimension scores, Italy-OECD comparison, and a simple regression model. Results show that social connections have the strongest dimension-level association with life satisfaction ($r = 0.453$), followed by environment and safety ($r = 0.305$) and work quality ($r = 0.300$). Italy reports lower life satisfaction than the OECD benchmark and performs particularly below it in social connections, civic engagement, work quality, and environment and safety.

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1 Research Question

1.1 Statement

Research question. Which dimensions of well-being are most associated with life satisfaction across OECD countries, and how does Italy compare with the OECD average excluding Italy?

The dependent variable is *life satisfaction*, measured on a 0–10 scale. The explanatory variables are not treated as isolated predictors only; instead, they are grouped into conceptual dimensions of well-being. This makes the analysis more interpretable and more consistent with the purpose of the project: starting from a relevant social question and using data analysis to answer it in a transparent and realistic way.

The explanatory dimensions are economic security, work quality, housing, health, social connections, civic engagement, and environment and safety. The aim is not only to check whether richer countries are more satisfied, but also to investigate whether broader dimensions of daily life are associated with subjective well-being.

1.2 Motivation and Stakeholders

The OECD argues that measuring progress requires looking beyond GDP and considering broader economic, social, and environmental outcomes for people [2]. This is the main motivation of the study: income and production are important, but they do not fully describe whether people live well.

The question is relevant for several stakeholders. Policy makers and governments may use well-being evidence to identify whether priorities should focus more on income, employment, health, housing, social relations, safety, or civic participation. Students and young workers are especially affected by employment quality, housing affordability, and social support. Researchers can use the results as a descriptive comparison of OECD countries. Italy is used as a country focus because it provides a concrete interpretation of the OECD benchmark.

This project is exploratory. It does not claim to prove that one variable causes higher or lower life satisfaction. The objective is to describe the strongest associations and discuss them carefully, especially given the limitations of aggregate country-level data.

2 The Data

2.1 Data Description

The dataset used in this project is the *OECD Well-being Data Monitor – Current well-being dataset*. The Current well-being data explorer allows users to explore measures of current well-being by country and compare them across OECD members [3]. The OECD has 38 member countries, which is also the number of countries selected in this project [4].

The original downloaded file contained 8,594 observations and 30 columns. After cleaning, filtering, and selecting the latest available observation for each country and measure, the final analytical dataset contained 38 OECD countries and 21 numeric well-being indicators plus the country name. Therefore, the final wide dataset had 38 rows and 22 columns.

Table 1: *Dataset filters used in the OECD Data Explorer.*

Filter	Selection
Time period	2010–2025
Reference area	38 OECD countries
Measure	21 well-being measures
Age	Total
Sex	Total
Education level	Total
Observation used	Latest available value

The variable *life satisfaction* is the outcome variable. The remaining variables are grouped into seven dimensions of well-being. The grouping is used to make the interpretation clearer and to avoid reducing the analysis to a list of isolated indicators.

Table 2: *Selected measures and conceptual well-being dimensions.*

Dimension	Selected measures
Outcome	Life satisfaction
Economic security	Households and NPISHs net adjusted disposable income per capita; median net wealth; household disposable income below the relative income poverty line
Work quality	Employment rate; labour market insecurity; long hours in paid work; job strain
Housing	Housing affordability; housing cost overburden; households living in overcrowded conditions
Health	Life expectancy at birth; perceived health as positive
Social connections	Social support; lack of social support
Civic engagement	Having a say in government; voter turnout
Environment and safety	Exposed to air pollution; exposure to extreme temperature; feeling safe at night; road deaths

2.2 Strengths and Weaknesses of the Data

The main strength of the dataset is that it comes from the OECD and is designed for cross-country comparison. It covers several dimensions of well-being, making it appropriate for a multidimensional analysis rather than a purely economic one. The OECD's *How's Life? 2024* report also emphasizes the use of many indicators to assess current well-being outcomes, inequalities, and resources for future well-being [5].

There are also limitations. First, not all variables are available for all countries. Second, different indicators may refer to different latest years. Third, the analysis is based on country-level aggregates and cannot explain individual differences within countries. Finally, the associations observed in the data should not be interpreted as causal effects.

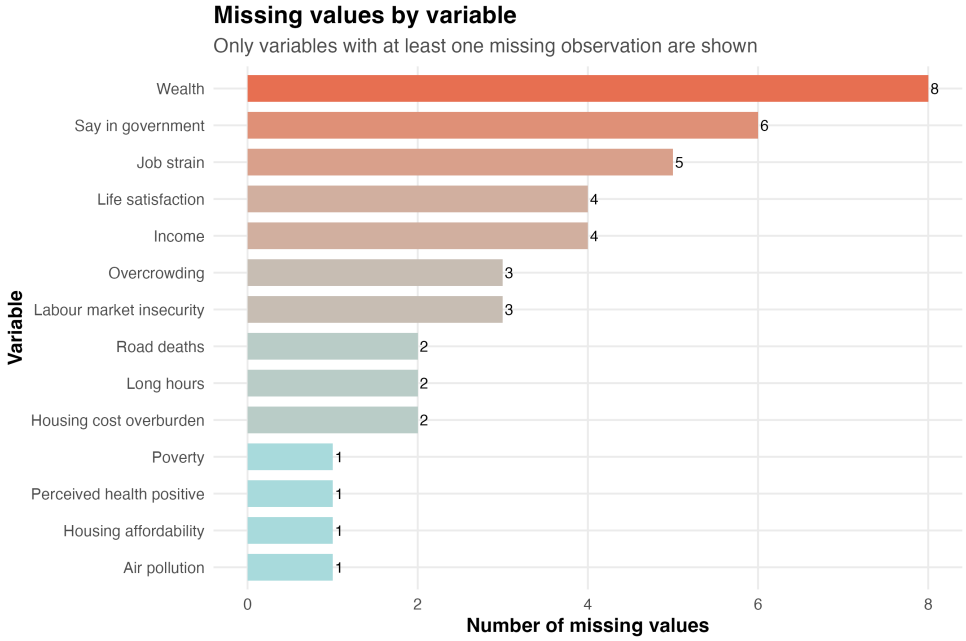


Figure 1: *Missing values for the selected indicators. The plot highlights which variables have the weakest data availability.*

3 Methodology

3.1 Overall Logic Thread of the Study

The methodology is organized as a continuous analytical workflow. The data are first cleaned and reshaped from long format to wide format, so that each country has one row and each selected indicator has its own column. After this preparation, the analysis examines descriptive statistics and missing values to understand the structure and quality of the data. The selected indicators are then correlated with life satisfaction to identify the strongest individual associations. Next, the indicators are grouped into broader well-being dimensions and transformed into standardized dimension scores, allowing variables measured in different units to be compared more consistently. Finally, Italy is compared with the OECD average (excluding Italy), and a simple regression model is estimated using the dimensions most strongly associated with life satisfaction.

Because the dataset covers 2010–2025 and not every indicator is available for every country in every year, using a single reference year would have produced many missing values. For this reason, the analysis selects the latest available observation for each country and each measure. This maximizes country coverage while keeping the analysis as recent as possible.

Therefore, the analysis should be interpreted as a latest-available cross-sectional comparison, not as a comparison based on one single common year for all indicators.

Listing 1: *Selection of latest available observation.*

```
df_latest_long <- df_selected %>%
  group_by(country, measure_key) %>%
  arrange(desc(year), .by_group = TRUE) %>%
  slice(1) %>%
  ungroup()
```

3.2 Tools: Which and Why

The analysis was conducted in RStudio. R was selected because it is appropriate for reproducible data cleaning, descriptive statistics, visualization, correlation analysis, and regression modelling. The main packages used were `tidyverse`, `janitor`, `skimr`, `broom`, and `ggplot2`.

No global imputation was applied. Descriptive statistics and correlations were computed using available observations. The regression model was estimated on complete cases for the selected variables. This choice avoids creating artificial values in a dataset with only 38 countries and keeps the analysis transparent.

3.2.1 Correlation Analysis

For each numeric indicator X_j, the Pearson correlation with life satisfaction Y is computed as:

$$r_{Y,X_j} = \frac{\sum_i (Y_i - \bar{Y})(X_{ij} - \bar{X}_j)}{\sqrt{\sum_i (Y_i - \bar{Y})^2}\sqrt{\sum_i (X_{ij} - \bar{X}_j)^2}}.$$

Positive values indicate that higher values of an indicator are associated with higher life satisfaction, while negative values indicate the opposite.

3.2.2 Dimension Scores

Since the variables are expressed in different units, each measure was standardized using a z-score:

$$z_{ij} = \frac{x_{ij} - \bar{x}_j}{sd_j}.$$

For indicators where higher values represent worse conditions, such as poverty, job strain, lack of social support, air pollution, road deaths, and housing cost overburden, the standardized value was multiplied by -1. Therefore, higher dimension scores always represent better conditions. The score for each dimension was then computed as the average of the available standardized components.

3.2.3 Regression Model

A simple regression model was used as an interpretative tool. The three dimensions with the strongest absolute correlations with life satisfaction were included: social connections, environment and safety, and work quality. The estimated model is:

$$LS_i = \beta_0 + \beta_1 SC_i + \beta_2 ES_i + \beta_3 WQ_i + \epsilon_i.$$

SC_i denotes the social-connections score, ES_i the environment-and-safety score, and WQ_i the work-quality score. The model is used to check whether the strongest dimensions remain informative when considered together.

4 Results

4.1 Steps of the Study: Description and Essential Numbers

Table 3 presents descriptive statistics for selected indicators. Across the OECD countries with available data, average life satisfaction is 7.33 on a 0–10 scale, with a standard deviation of 0.54. The observed values range from 5.70 to 8.45. This shows that most OECD countries are concentrated within a relatively narrow range of life satisfaction, although relevant differences between countries still exist.

Table 3: *Descriptive statistics for selected indicators.*

Indicator	n	Mean	SD	Min	Max
Life satisfaction	34	7.33	0.54	5.70	8.45
Employment rate	38	78.72	5.65	59.24	87.61
Social support	38	89.83	3.85	79.82	97.66
Life expectancy	38	81.13	2.50	75.50	84.30
Housing cost overburden	36	18.26	10.13	3.76	48.53
Relative income poverty	37	11.48	3.90	5.52	21.02
Job strain	33	27.59	6.87	13.83	47.94
Perceived health positive	37	68.62	10.96	37.10	85.60
Road deaths	36	4.61	2.21	1.45	12.24

The first analytical step was to compute correlations between life satisfaction and the selected individual measures. Table 4 reports the strongest individual associations.

Table 4: *Strongest individual correlations with life satisfaction.*

Measure	Dimension	Correlation
job strain	Work quality	-0.541
perceived positive health	Health	0.516
road deaths	Environment and safety	-0.494
social support	Social connections	0.461
relative income poverty	Economic security	-0.457
lack of social support	Social connections	-0.442
labour-market insecurity	Work quality	-0.350
employment rate	Work quality	0.297
air pollution exposure	Environment and safety	-0.289
extreme temperature exposure	Environment and safety	-0.184
say in government	Civic engagement	0.181
disposable income	Economic security	0.089

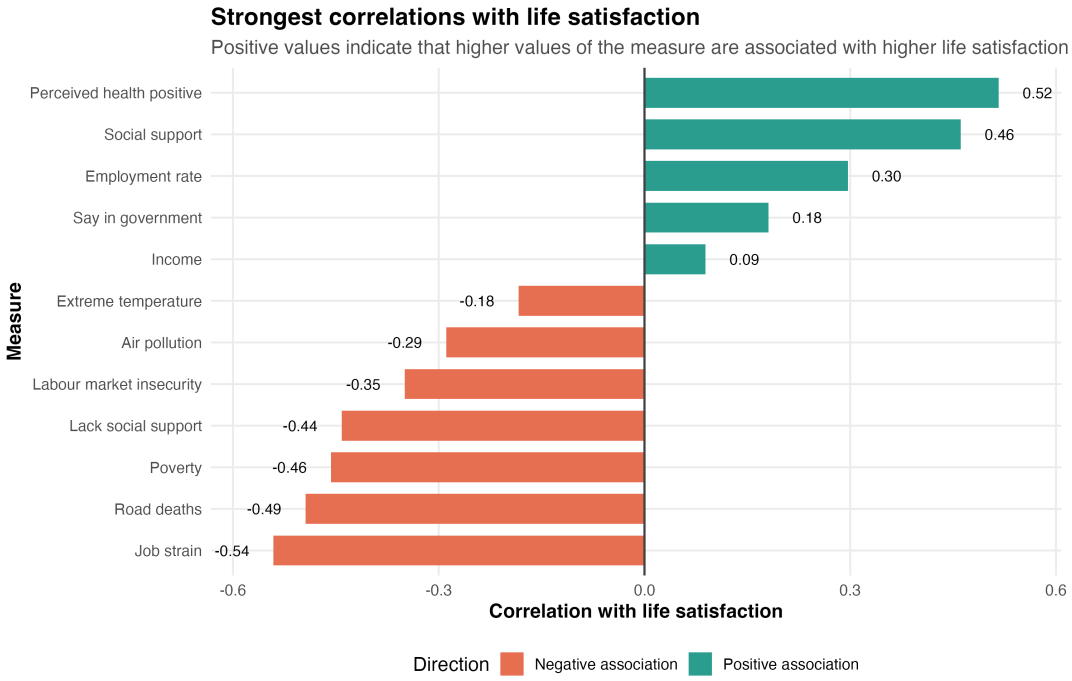


Figure 2: *Strongest individual correlations with life satisfaction. Teal bars indicate positive associations; red-orange bars indicate negative associations.*

At the dimension level, social connections show the strongest association with life satisfaction, with a correlation of 0.453. Environment and safety have the second strongest association, with a correlation of 0.305, followed by work quality with 0.300. Economic security also shows a positive relationship, but weaker than social connections, with a correlation of 0.245.

Table 5: *Dimension-level correlations with life satisfaction.*

Dimension	Correlation
Social connections	0.453
Environment and safety	0.305
Work quality	0.300
Economic security	0.245
Health	0.162
Housing	-0.069
Civic engagement	-0.008

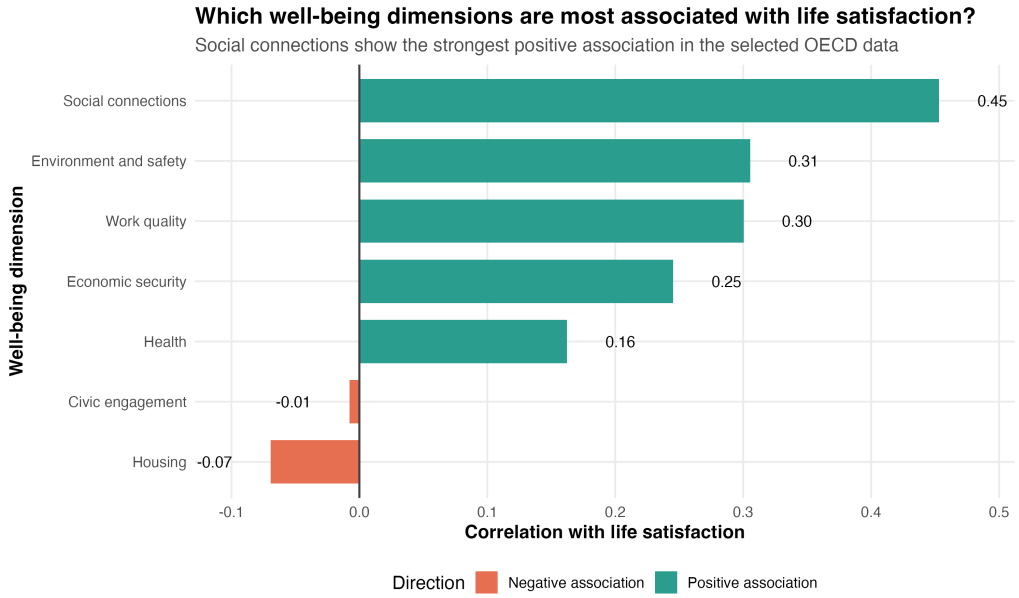


Figure 3: *Correlation between standardized well-being dimension scores and life satisfaction.*

Job strain is the strongest individual correlate of life satisfaction [Figure 2]. Figure 4 shows the country-level relationship between job strain and life satisfaction. The fitted line slopes downward, confirming the negative association.

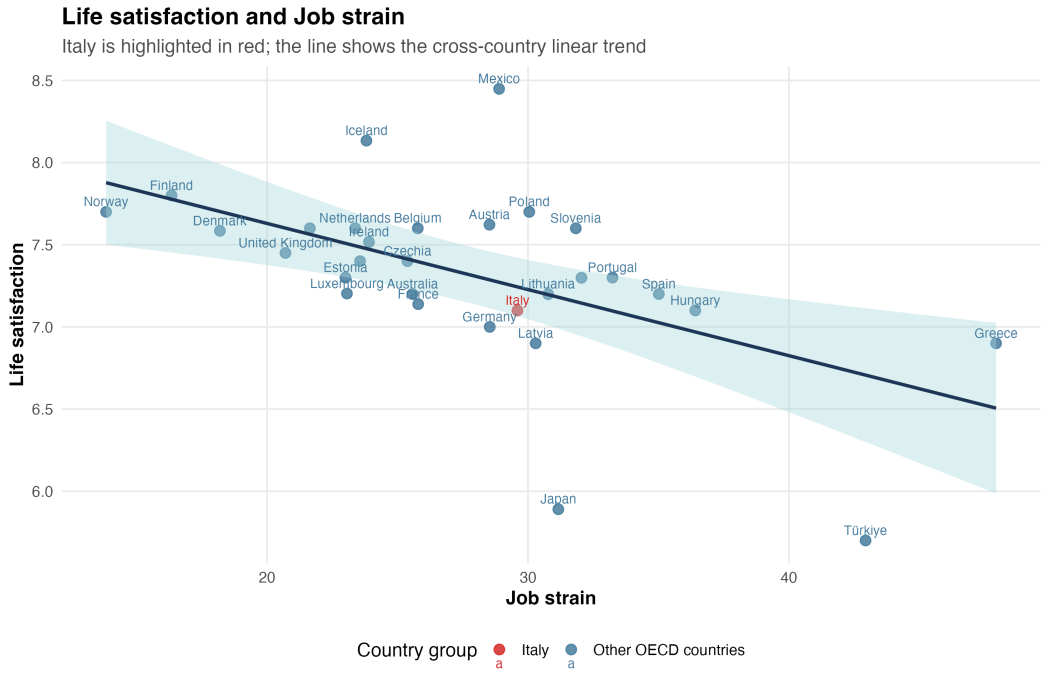


Figure 4: *Scatterplot of life satisfaction and job strain. The fitted regression line indicates a negative association.*

Italy is compared with the OECD average (excluding Italy). This choice avoids including Italy in the benchmark against which it is evaluated. Since dimension scores are standardized, values above zero indicate better-than-average conditions relative to the OECD distribution, while values below zero indicate weaker conditions.

Table 6: *Italy compared with the OECD average (excluding Italy).*

Indicator	Italy	OECD excl. Italy	Difference
Life satisfaction	7.100	7.334	-0.234
Health	0.784	-0.052	0.835
Economic security	0.026	-0.059	0.086
Housing	-0.158	-0.005	-0.154
Environment and safety	-0.295	0.000	-0.295
Work quality	-0.517	-0.019	-0.498
Civic engagement	-0.657	0.018	-0.675
Social connections	-0.935	0.025	-0.960

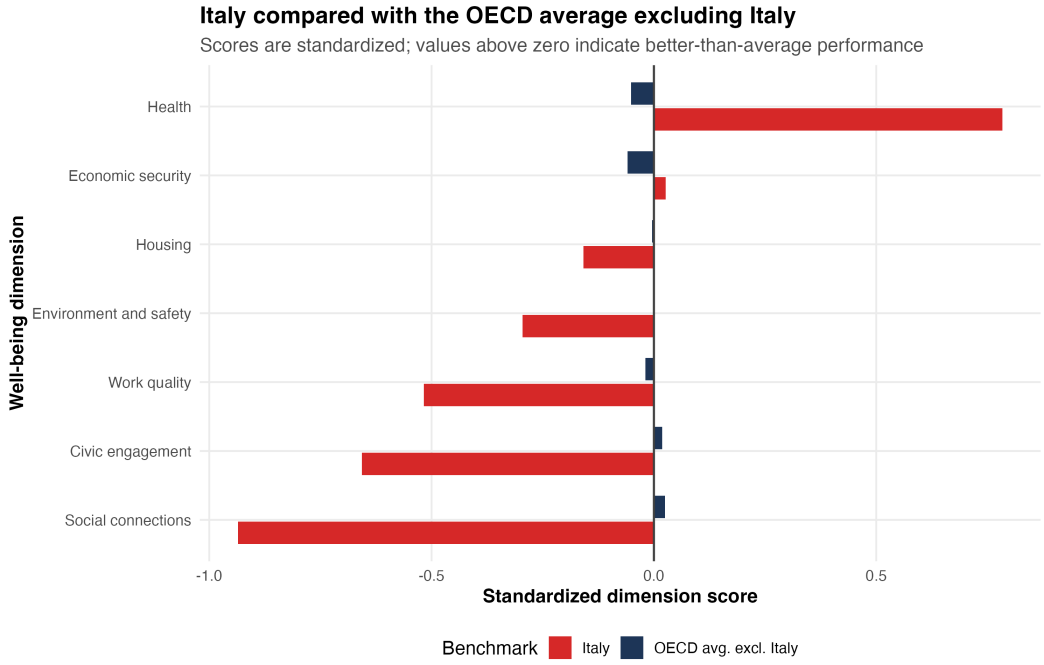


Figure 5: *Italy compared with the OECD average (excluding Italy) across standardized well-being dimensions. Positive values indicate better-than-average performance relative to the OECD distribution, while negative values indicate weaker performance. The blue bars represent the OECD average (excluding Italy).*

Finally, a regression model was estimated using the three strongest dimensions from the correlation analysis. The model uses 34 complete observations because four countries are missing life satisfaction values or some variables used in the model.

Table 7: *Regression coefficients. Dependent variable: life satisfaction.*

Term	Estimate	SE	t	p-value
Intercept	7.321	0.089	82.286	0.000
Social connections	0.229	0.114	2.005	0.054
Environment and safety	0.059	0.253	0.232	0.818
Work quality	-0.035	0.208	-0.166	0.869

Table 8: *Regression model summary.*

Statistic	Value
Observations	34
R^2	0.207
Adjusted R^2	0.127
Residual standard error	0.501
F-statistic	2.606
Model p-value	0.070

4.2 Comments and Interpretation

The results tells us that among the well-being dimensions, social connections are most strongly associated with life satisfaction. This finding is important because it shows that subjective well-being is not simply a function of income or wealth. Economic security is positively related to life satisfaction, but its dimension-level correlation is weaker than that of social connections.

The second group of relevant dimensions is environment and safety and work quality. These dimensions are also positively associated with life satisfaction when bad indicators are reversed. At the individual-measure level, job strain is particularly important, which suggests that working conditions can influence well-being beyond the simple employment rate.

Housing and civic engagement show weak associations in this dataset. This does not imply that housing and civic engagement are unimportant in general. Instead, it suggests that the selected indicators, the latest-observation design, and the small cross-country sample do not reveal a strong linear relationship with life satisfaction for those dimensions.

The Italy comparison strengthens the interpretation. Italy is relatively strong in health, close to the benchmark in economic security, but weaker in social connections, civic engagement, work quality, environment and safety, and housing. Since social connections are the strongest dimension-level correlate of life satisfaction, Italy's low social-connection score is especially relevant.

The regression explains about 20.7% of the cross-country variation in life satisfaction. The coefficient for social connections is positive and close to conventional statistical significance ($p = 0.054$). The coefficients for environment and safety and work quality are not statistically significant in the joint model. Therefore, the regression should be interpreted as supporting evidence, not as a definitive explanatory model.

The small negative coefficient for work quality should not be interpreted as evidence that better work quality reduces life satisfaction. In the simple dimension-level correlation, work quality is

positively associated with life satisfaction, while the sign changes only in the joint regression, likely because the dimensions overlap and the sample contains only 34 complete observations.

5 Conclusion

This project analysed which dimensions of well-being are most associated with life satisfaction across OECD countries and compared Italy with the OECD average (excluding Italy). The analysis used the OECD Well-being Data Monitor, selecting 21 measures for 38 countries over the period 2010–2025 and retaining the latest available observation for each country and measure.

The main result is that social connections have the strongest dimension-level association with life satisfaction. Environment and safety and work quality follow, while economic security is positively associated but not dominant. At the individual-measure level, job strain, perceived positive health, road deaths, social support, poverty, and lack of social support are among the strongest correlates.

For Italy, the analysis shows a lower life satisfaction score than the OECD average (excluding Italy). Italy's main strength is health, while its main weaknesses are social connections, civic engagement, work quality, and environment and safety. These results suggest that policies aimed at improving quality of life should not be limited to income growth, but should also address social support, job strain, safety, and broader living conditions.

The main strength of the study is its use of internationally comparable OECD data and an interpretable methodology. The main weakness is that the analysis is based on aggregate country-level data, which prevents individual-level conclusions and causal claims. Future research could improve the analysis by using individual-level survey data, adding a real time-series component, or focusing on specific groups such as young people, low-income households, or workers.

6 References

References

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[2] OECD. *Well-being and beyond GDP*. Available at: https://www.oecd.org/en/topics/policy-issues/well-being-and-beyond-gdp.html

[3] OECD. *Current well-being Data Explorer*. Available at: https://data-explorer.oecd.org/vis?fs[0]=Topic,1%7CSociety%23SOC%23%7CWell-being%20and%20beyond%20GDP%23SOC_WEL%23&pg=0&fc=Topic&bp=true&snb=26&df[ds]=dsDisseminateFinalDMZ&df[id]=DSD_CWB%40DF_CWB&df[ag]=OECD.WISE.CWB&df[vs]=1.0&dq=..A1&lom=LASTNPERIODS&lo=5&to[TIME_PERIOD]=false.

[4] OECD. *Members and partners*. Available at: https://www.oecd.org/en/about/members-partners.html.

[5] OECD. *How's Life? 2024: Well-being and Resilience in Times of Crisis*. OECD Publishing, Paris, 2024. Available at: https://www.oecd.org/en/publications/2024/11/how-s-life-2024_bdcf2f9f.html.

A Appendix

A.1 Variables Description

Table 9: *Complete list of selected variables and dimensions.*

Variable	Dimension	Interpretation
life satisfaction	Outcome	Average evaluation of life on a 0–10 scale
income	Economic security	Disposable income per capita
wealth	Economic security	Median net wealth
poverty	Economic security	Share below relative income poverty line; higher is worse
employment rate	Work quality	Share of working-age population employed
labour-market insecurity	Work quality	Expected earnings loss from unemployment; higher is worse
long hours	Work quality	Share working long paid hours; higher is worse
job strain	Work quality	Share experiencing high job strain; higher is worse
housing affordability	Housing	Housing affordability indicator; higher is better
housing cost overburden	Housing	Housing cost overburden; higher is worse
overcrowding	Housing	Share living in overcrowded conditions; higher is worse
life expectancy	Health	Life expectancy at birth
perceived health positive	Health	Share reporting positive health
social support	Social connections	Share reporting social support
lack social support	Social connections	Share lacking social support; higher is worse
say in government	Civic engagement	Share reporting a say in government
voter turnout	Civic engagement	Voter turnout
air pollution	Environment and safety	Share exposed to air pollution; higher is worse
extreme temperature	Environment and safety	Exposure to extreme temperature; higher is worse
feeling safe	Environment and safety	Share feeling safe at night
road deaths	Environment and safety	Road deaths; higher is worse

A.2 Full Descriptive Statistics

Table 10: *Descriptive statistics for all numeric indicators.*

Variable	n	Mean	SD	Min	Max
air pollution	37	94.95	11.71	47.85	100.00
employment rate	38	78.72	5.65	59.24	87.61
extreme temperature	38	6.52	13.23	0.00	44.19
feeling safe	38	72.99	12.47	37.40	90.73
housing affordability	37	80.14	2.86	73.76	85.63
housing cost overburden	36	18.26	10.13	3.76	48.53
income	34	39811.26	9576.36	19128.00	63190.00
job strain	33	27.59	6.87	13.83	47.94
labour-market insecurity	35	5.11	4.26	0.99	21.68
lack social support	38	9.63	3.88	2.17	19.92
life expectancy	38	81.13	2.50	75.50	84.30
life satisfaction	34	7.33	0.54	5.70	8.45
long hours	36	6.47	5.98	0.49	25.91
perceived health positive	37	68.62	10.96	37.10	85.60
poverty	37	11.48	3.90	5.52	21.02
road deaths	36	4.61	2.21	1.45	12.24
say in government	32	29.43	10.68	12.10	58.17
social support	38	89.83	3.85	79.82	97.66
voter turnout	38	70.95	11.02	46.64	90.67
wealth	30	217036.87	141907.85	61894.00	853921.00
overcrowding	35	11.53	8.64	0.59	31.57

A.3 Reproducibility Code Excerpt

Listing 2: *Core R workflow used for the analysis.*

```r
library(tidyverse)
library(janitor)
library(broom)

# Import data
df_raw <- read_csv("OECD_WELLBEING_DATA.csv", show_col_types = FALSE) %>%
  clean_names()

# Latest observation for each country and measure
df_latest_long <- df_selected %>%
  group_by(country, measure_key) %>%
  arrange(desc(year), .by_group = TRUE) %>%
  slice(1) %>%
  ungroup()

# Wide dataset for correlations and regression
df_wide <- df_latest_long %>%
  select(country, measure_key, value) %>%
  pivot_wider(names_from = measure_key, values_from = value)

# Italy vs OECD benchmark (excluding Italy)
```

```r
oecd_average_row <- analysis_scores %>%
  filter(str_to_lower(country) != "italy") %>%
  summarise(across(where(is.numeric), ~ mean(., na.rm = TRUE)))

# Regression with the three strongest dimensions
model <- lm(life_satisfaction ~ dim_social_connections +
              dim_environment_safety + dim_work_quality,
            data = analysis_scores)

tidy(model)
glance(model)
```